



An Energy Efficient and Reliable Routing Scheme to enhance the stability period in Wireless Body Area Networks

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ABSTRACT

Wireless Body Area Network (WBAN) is a wireless network of wearable sensing and computing devices connected through a wireless communication channel, thereby offering a plethora of enticing applications in the area of remote health monitoring, sports, and entertainment. However, WBANs nodes are highly resource-constrained; therefore, energy-efficient and reliable data transmission is very vital in the design and implementation of most of its applications. Furthermore, on time and accurate data delivery with minimum delay is also highly required. Over time, numerous energy-efficient routing solutions have been proposed for WBANs; however, the significant feature of reliability in these solutions has not been adequately addressed. Therefore, in this paper, we propose a new Energy-Efficient and Reliable Routing Scheme (ERRS) to enhance the stability period and reliability for resource-constrained WBAN. ERRS comprises two novel solutions, namely, the Forwarder Node Selection and Forwarder Node Rotation techniques. The proposed ERRS takes advantage of the adaptive static clustering routing technique and achieves enhanced stability period and longer network lifetime, ultimately maximizing reliability. Through extensive simulation-based evaluation using MATLAB, ERRS showed an improvement of 26% over the benchmark protocol in terms of network stability and throughput. Whereas the end-to-end delay of the proposed ERRS is improved by 17% and 40% than by SIMPLE and M-ATTEMPT protocols, respectively, which proves ERRS to be an efficient and reliable routing solution for WBANs.

1. Introduction

Currently, due to the growth in the aging population and scarce financial resources, traditional health care systems are facing many challenges. According to an estimate of the Committee for a Responsible Federal Budget, in the US, National healthcare spending is projected to grow to \$3.6 trillion in 2019 and is anticipated to rise in the coming years. In Western countries, elderly and retired parents usually do not live with their children and are admitted to the establishment or Care Centers that provide living quarters and health care for elderly citizens or patients. A system that can continuously monitor the health condition of older people and patients and share information with remote care providers or hospitals are highly demanding. Therefore, scientists and researchers are being motivated to minimize the increasing cost of remote medical health care and patient health monitoring with optimum and affordable solutions.

To catch-up with these trends, Wireless Body Area Network (WBAN) has recently emerged as a new trend and attractive technology to

provide such viable healthcare solutions. Technically, WBAN is a group of interconnected, low power, smart, and miniaturized sensor-nodes, which communicate their sensed data through a wireless medium in or around the human/patient body [1]. WBAN is used to measure physical parameters in order to monitor any critical or abnormal conditions in the human body [2]. These parameters include Electro-Cardiograph (ECG), which measures the electrical activity of the heart, Electro-Encephalography (EEG), which measures the electric activity of the brain, and Electromyography (EMG) which measures the electrical activity of body muscles. Subsequently, the collected/aggregated information is transferred to control devices such as Sink or Gateway for further analysis and diagnostics [3].

In principle, WBAN is a subset of Wireless Sensor Networks (WSNs), while WSN is a set of interconnected sensor-nodes having the capability of sensing, processing, and communication through a wireless medium (i.e., Wi-Fi, Bluetooth, and Zig-Bee) [4]. In terms of implementation and network architecture, WBAN is classified into two types,

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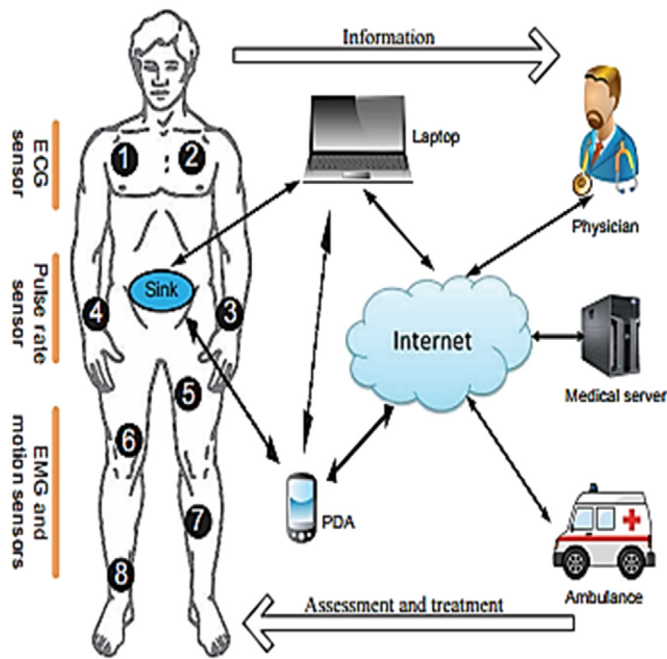


Fig. 1. Monitoring of the patient body [10].

i.e., Implanted/In-Body WBAN and Wearable/On-Body WBAN [5]. The former means that implanted bio-sensors or Sink node can transfer information with each other, whereas the latter means the sensor nodes which are placed on the human body, sense vital parameters and transmit this information to the Sink node [6]. Similarly, each sensor in WBAN is broadly classified into three categories, such as physiological, bio-kinetic, and ambient sensors [6]. Physiological sensors measure physical phenomena/attributes such as blood pressure, temperature, and monitoring of glucose, etc. Bio-kinetic sensors can measure acceleration and angular rate of rotation of movement of a human body while the sensors which can measure any environmental conditions, i.e., light, vibration and pressure level, etc. are called ambient sensors [7], as shown in Fig. 1. WBANs have a plethora of applications, ranging from both medical and non-medical applications [8,9]. Medical applications of WBANs are divided into wearable and implanted medical applications such as *Sleep Staging*, *Asthma*, *Sports*, *Battlefield*, *Diabetes Control*, and *Cardiovascular Monitoring* applications [1]. The Non-Medical applications of WBANs are used in the *Entertainment Sector*, *Consumer Electronics Sector*, *Off-Body (built into the house)*, *Lifestyle Sector*, and in *aerospace*.

Due to the resource-constrained nature of WBAN (such as limited power supply, memory, bandwidth, and processing capabilities) and wireless communication channels, there are many issues and challenges associated with the design and implementation of WBANs [11]. Issues related to energy-efficiency and reliability are among the most important ones and challenging ones at the same time. It is due to the fact that energy-efficiency and reliability has a great impact on the overall lifetime and stability period of the network and application [3]. In the context of WBAN, reliability implies that the sensitive data may transfer to the sink node in a timely and correct manner while stability period means the total time of network operations till the death/expiry of the first sensor node. Since, in WBAN, sensor nodes are operated on batteries and in many applications' scenario, recharging or replacing of batteries is almost impossible, it is important to save the battery power on the face of continuous monitoring [11]. There is a need for energy-efficient and reliable routing solutions, which may enhance the stability period and maximize the overall network lifetime so that the aim of implementing a WBAN application is achieved.

In WBAN, energy-efficient and reliable routing protocols are classified into cross-layer, thermal aware, clustered based, delay-tolerant, and quality of service (QoS) based routing [1,12]. These solutions are either based on a single hop or multi-hop communication inspired by WSNs. Cross-layer routing protocols are considered unsuitable for WBAN because loads are not uniformly distributed among sensor-nodes, which can increase the chances of energy holes [13]. In addition, for tiny resource-constrained sensor-nodes, cross-layer protocols incur excessive computational overhead with increased complexity. On the other hand, thermal-based approaches are widely used for implanted WBAN [13]. However, the detection of hotspot nodes incurs more overhead and consumes high energy in wearable WBAN. Similarly, clustering-based routing approaches proved to be more suitable for WBANs applications as compare to cross-layer and thermal-aware routing approaches [14].

In the existing literature, clustering-based routing approaches are mainly used to increase the energy-efficiency and enhance reliability to achieve longevity in network operations of WBANs' applications. However, there are still some important issues and challenges that need considerable attention. For instance, in many solutions, energy conservation during the network deployment, network organization, and inter-nodes communication phases have been ignored [11]. Similarly, in [15], the selection criterion of a forwarder node and the optimum number of relay nodes is unsuitable, where the load is not evenly distributed among sensor-nodes. Moreover, a round by round selection of the forwarder node is not optimal in terms of energy consumption, reliability, and maximum throughput in some solutions [16–18]. To resolve these issues, a new energy-efficient and reliable routing scheme, known as ERRS (Energy Efficient and Reliable Routing Scheme) is proposed in this paper. The proposed ERRS is based on an adaptive static clustering approach that exploits both the advantages of static as well as dynamic clustering to conserve energy and enhance the reliability of WBANs.

Furthermore, ERRS is comprised of two novel solutions, namely, the Forwarder Node Selection and Forwarder Node Rotation techniques. The proposed ERRS takes advantage of the adaptive static clustering routing technique to achieve an enhanced stability period and a longer network lifetime. From the architectural and system design standpoint, the overall operations of the proposed ERRS are divided into four phases, i.e., Network Deployment and Initialization, Forwarder Node Selection, Data Transmission, and Load-Balancing. The first three phases are required for the initial operations of data gathering of the deployed application; however, the system turns into a load-balancing phase when several data gathering rounds reach to a pre-defined threshold value.

The contribution of this paper is two-fold, i.e.:

- First, a novel technique is proposed for the selection of the forwarder node in WBANs to increase energy-efficiency and network stability period.
- Second, a novel technique is proposed to rotate the role of a forwarder node among the deployed sensor-nodes to ensure even and uniform load-balancing, hence all the sensor-nodes get a chance to become a forwarder node. As a result, the technique reduces energy consumption and extends the network lifetime, thereby enhancing the reliability of WBANs.

The rest of the paper is organized as follows: Section 2 consists of related work proposed in the context of this paper. In Section 3, the proposed ERRS scheme is presented in detail. In Section 4, we discuss simulation results and analysis, while Section 5 gives the conclusion along with future work.

2. Related work

In this section, we critically analyze the existing solutions summing up their strengths and weaknesses. After the in-depth literature review, we concluded that the energy-efficient and reliable routing solutions for WBAN could be categorized into various classes, i.e., *Cross-layer*, *Thermal aware*, *Cluster-based*, *Quality of service*, and *Delay-tolerant aware* routing approaches based on their underlying mechanism and system [12,19–21].

In these solutions, Cross-layer routing protocols are found unsuitable for WBANs. Because, in cross-layer protocols, the traffic load is not uniformly distributed among sensor-nodes, which ultimately increases the chances of energy holes. Besides, for tiny resource-constrained sensors, cross-layer protocols incur excessive computational overhead with increased complexity [12,22]. On the other hand, thermal-based approaches are widely used for implanted WBANs. Moreover, the detection of hotspot nodes incurs more overhead and consumes high energy during network operations [13,22]. However, clustering-based routing solutions proved to be more suitable for WBANs applications as compared to cross-layer and thermal-aware routing approaches due to their intrinsic advantages of load-balancing and energy conservation [12,14]. In a cluster-based routing approach, data is sent to the sink node through cluster heads/forwarder nodes in their allocated time slots, which minimizes energy consumption with high throughput hence increases stability period [11,14]. Furthermore, cluster-based approaches are used to enhance the network lifetime by reducing energy consumption in data aggregation and dissemination.

Javaid et al. [23] proposed a well-known thermal aware routing protocol called as M-ATTEMPT (Mobility Supporting Adaptive Threshold Based Thermal Aware Energy-Efficient Multi-Hop Protocol). In this approach, while routing data must be routed away from those sensor nodes which can trace/sense link hotspot. The shortest path is selected among various accessible paths. If several routes have a similar number of hops, a path with low energy depletion to sink is selected. A joint-invitation request is sent if a sensor node changes its position during a data gathering round, and a joint-request is accepted when the number of child nodes is less than three (03) after checking the child's list of the parent nodes. Packets are sent on a long alternate path when sensor nodes sense links hotspot, which leads to higher energy consumption. Albeit of its advantages, in M-ATTEMPT, there is no other path that exists when a sensor dies. Furthermore, these sensor nodes are located according to their data rate instead of their energy level. To overcome the limitation of M-ATTEMPT, Ahmad et al. [24] proposed the concept of a thermal un-aware routing approach known as RE-ATTEMPT (Reliability Enhanced Adaptive Threshold Based Thermal Un-Aware Energy-Efficient Multi-Hop Protocol). RE-ATTEMPT adopts a multi-hop communication paradigm where the shortest path is selected among sensors and sink nodes. In the RE-ATTEMPT protocol, sensor nodes find all the available routes through the exchange of HELLO messages. When a current path is not accessible, it selects an alternate sub-optimal path to ensure reliable transmission. This approach can avoid relaying of un-necessary traffic due to the limited number of forwarder nodes. However, It is not the best choice to confirm low latency at the cost of energy and reliability because direct communication with the sink node will increase the chances of packet loss due to long-distance [25].

Tauqir et al. [8] proposed an energy-efficient and a high throughput clustered-based routing protocol known as Distance Aware Relaying Energy-Efficient Protocol (DARE) for WBANs. Using DARE, eight patients were monitored in a medical ward by considering different scenarios for the placement of the sink. In their implementation, seven sensor nodes were deployed/attached with each patient, including one relay node on the chest and one main sensor on the bed, with unlimited energy resources. DARE builds a heterogeneous network consisting of a main sensor and body relay nodes by using the multi-hop technique. In DARE, the propagation delay is found comparatively very high. Furthermore, link quality, path loss, and network cost were not properly considered [26].

Nadeem et al. [27] proposed an energy-efficient routing protocol known as SIMPLE (Stable Increased Throughput Multi-Hop Protocol for Link Efficiency). In SIMPLE, a forwarder node is selected based on the least-cost function. In each round, those nodes are selected as a forwarder node that has higher residual energy and minimum distance to the sink node. In SIMPLE, the residual energy parameter balances the energy depletion among sensor-nodes, while the distance parameter ensures maximized throughput. In this scheme, the network lifetime, stability period, and throughput are increased as compared to M-ATTEMPT protocol. However, the criterion to select a forwarder node is inapt because traffic load is not equally distributed among sensor-nodes. Moreover, in critical data reporting-based applications, the performance of the SIMPLE protocol degrades due to the continuous data transmission because it does not compare current sensed value and previously sensed value. An extension to SIMPLE, Javaid et al. [28] proposed the IM-SIMPLE (Improved Stable Increased Throughput Multi-Hop Protocol for Link Efficiency) protocol for WBAN to overcome the deficiencies exist in SIMPLE protocol. The main goal of the IM-SIMPLE protocol was to handle mobility. Furthermore, a linear programming model is used to maximize throughput. IM-SIMPLE used a multi-hop communication approach for normal sensor-nodes to save energy. Furthermore, a sink node can select a forwarder node with a minimum cost function, and all other sensor nodes can transmit data to the forwarder node in their allocated time slot by using TDMA scheduling mechanism. One of the main drawbacks of this approach is the poor selection criterion for the forwarder node. A node nearer to the sink is often selected as a forwarder node, and most of the time, the energy of such nodes depletes rapidly; hence they die soon. Furthermore, in IM-SIMPLE protocol, the number or percentage of packets dropped over a particular period is higher than SIMPLE since less attention is given to the link quality and path-loss [29].

Sandhu et al. [11] proposed the FEEL (Forwarding Data Energy-Efficiently with Load Balancing) protocol to increase energy-efficiency and reliability in WBANs. FEEL is an energy-efficient protocol as it consumes battery energy inefficient manner, which ultimately improves the network stability period. For the selection of a forwarder node, effective criteria have been specified as compared to SIMPLE and IM-SIMPLE protocols. This effective forwarder node selection ultimately results in maximizing throughput and network lifetime. However, less attention is given to the retransmission of data packets in case of path loss [20]. Similarly, in [30], the authors proposed an energy-efficient link aware routing scheme for WBAN known as LAEEBA (Linked Aware Energy-Efficient Routing Protocol). LAEEBA proved to be a reliable scheme with minimum path loss, and the overall throughput of the network is high. In addition, the network lifetime is recorded higher using both single-hop and multi-hop communications, which reduces the effects of path-loss. Furthermore, LAEEBA used the same criteria for forwarder node selection as used by SIMPLE protocol that proved to be efficient in terms of energy-efficiency and stability period of the network. However, The main drawback of this protocol is more overhead of data packets during data communication [29].

To overcome the deficiencies in LAEEBA, Ahmed et al. [31] proposed the CO-LAEEBA (Cooperative Linked Aware Energy-Efficient Routing Protocol for WBAN), which is an energy-efficient solution based on multicasting cooperative approach. In CO-LAEEBA, two or more paths exist in which critical data transmission is carried through a single-hop communication while normal data delivery follows a path via relay nodes (a node which acted as a cooperative node). Packets are directly delivered to the sink when the remaining energy of a sensor-node is greater than a relay node. When the remaining energy of a sensor node is less than a relay node, it follows the multi-hop communication. Simulation results of CO-LAEEBA show improved performance in terms of maximum throughput and increased network lifetime over the existing selected protocols. However, in CO-LAEEBA protocol, too many links are required for source to destination delivery of data packets because a limited number of the forwarder nodes are used [29].

Umer et al. [32] proposed the multi-hop Hybrid Rapid Response Routing Protocol (HRRR) for WBANs, which is used for delay-sensitive traffic where time-critical data is transmitted with minimum delay. This protocol is energy-efficient and delay-tolerant due to the two main features, i.e., its large coverage area and real-time transmission of patient data. QoS-based HRRR protocol not only favors the timely transmission of critical data of patients in WBANs but also conserves the energy. In contrast to other routing protocols of WBANs, HRRR protocol provides the QoS-aware energy efficiency, fault-tolerance, reliability, and robustness. However, the issues of load balancing and packet overhead are not properly addressed by the authors. Similarly, Mehmood et al. [33] adopted the cooperative communication and network coding technique to enhance fault-tolerance and maximize energy-efficiency in WBANs. By employing network coding and cooperative communication, they managed to reduce energy consumption with increased throughput. However, they used the cooperative communication without considering its potential adverse effects of increased processing overhead.

BEC (Balanced Energy Consumption) is another clustered-based routing protocol proposed by Sandhu et al. [15] in which a relay node is selected based on cost function depending on the residual energy of sensor nodes. Relay/Forwarder nodes are selected for each round based on the least cost function for the distribution of load uniformly. If the node is closer to the sink, it sends its data directly; otherwise, it is sent to the nearest relay node. The threshold value for residual energy is set; if it is met, the node sends only its critical data to the sink. The simulation study shows that this protocol works well in terms of network lifetime. However, the selected forwarder nodes expend more energy due to the vigorous task of forwarding [20]. Similarly, most recently, Ullah et al. [34] proposed a cluster-based dual sink node routing protocol (DSCB), where the concept of dual sink is used. The use of double sinks can improve the negative consequences of the shadow effect causes, balance the network load of a single sink node, and improve the practicability of the WBAN. However, it is highly required to address the energy-efficiency with nodes cooperation along with the energy harvesting for efficient routing. Kumar et al. [35] proposed a Lightweight QoS Routing Protocol (LRPD), which is designed to reduce delay. To achieve goal optimization, the LRPD protocol adopts a modular approach where data from the upper layer first enters the data classification module for priority division. Then data is classified into two types of packets, i.e., general packets (GP) and delay sensitive packets (DP). To make sure the real time transmission of delay sensitive data, the high priority data will not wait for direct transmission, while the low priority data needs to wait until an appropriate transmission time. However, extra computational overhead and higher transmission delay are the main noted drawbacks of this scheme.

Though, various energy-efficient and reliable clustered based routing approaches have been proposed to increase the network stability period, maximize throughput, and achieve longevity in network operations of WBANs. However, the important issue of reliability is not properly addressed. To address this problem, we propose an Energy-Efficient and Reliable Routing Scheme (ERRS), which is based on the adaptive static clustering approach that exploits the advantages of both static and dynamic clustering. The proposed scheme conserve not only energy but also enhances the stability period and reliability of WBANs.

3. The proposed Energy Efficient and Reliable Routing Scheme (ERRS)

In this section, we present our proposed approach, named as an Energy-Efficient and Reliable Routing Scheme (ERRS) for WBAN.

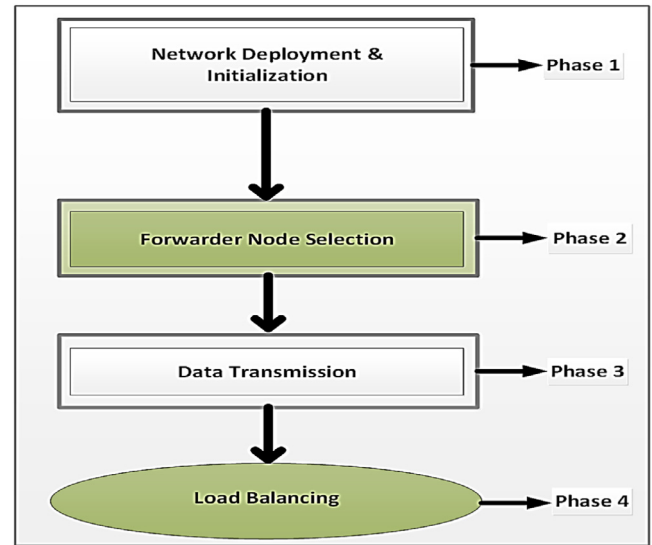


Fig. 2. WBAN network operations.

3.1. Motivation

In WBANs, an excessive amount of energy is consumed when sensor-nodes communicate or gather data as compared to mere sensing. Therefore, the limited sensor-node energy can affect the overall operations of WBANs applications, including the network lifetime and reliability [36, 37]. To address this challenge, several energy-efficient clustering based routing protocols have been proposed over the past to enhance the network lifetime; however, not as much attention is given to the aspect of reliability in terms of stability period of the network [11], as discussed in the previous section. For instance, most of the cluster-based schemes have vague criteria for the selection of forwarder nodes and having uneven distribution of traffic load among deployed sensor-nodes. Besides, a round by round selection of the forwarder node is not ideal in terms of energy consumption and reliability. Given that, the selection criteria of the forwarder node and traffic load distribution among the sensor-nodes are the most challenging task to eliminate the energy hole's problem in WBANs. When a traffic load is not uniformly distributed among the sensor-nodes, the stability period of the network decreases because sensor-nodes may deplete their energy and die quickly [11].

The second main issue in WBANs is reliability, which is related to two parameters, i.e., timely delivery of data and correctness of data when received at the destination. The lack of reliability of the network can directly affect the quality of patient monitoring because, in a worst-case scenario, it can be harmful when a life-threatening event has gone undetected. Our proposed approach, i.e., ERRS, not only ensures reliable and efficient transmission of data but also balances the energy consumption of the network, thereby prolonging the overall network lifetime and stability period.

3.2. System design overview of ERRS

From the architectural and system design standpoint, the overall operations of the proposed ERRS scheme are divided into four working phases, i.e., **Network Deployment and Initialization**, **Forwarder Node Selection**, **Data Transmission**, and **Load Balancing** as shown in Fig. 2. The first three phases are required for the initial operations of data gathering of the intended application (e.g., patient's monitoring), while the system turns into a load-balancing phase when several data gathering rounds reach a pre-defined threshold value.

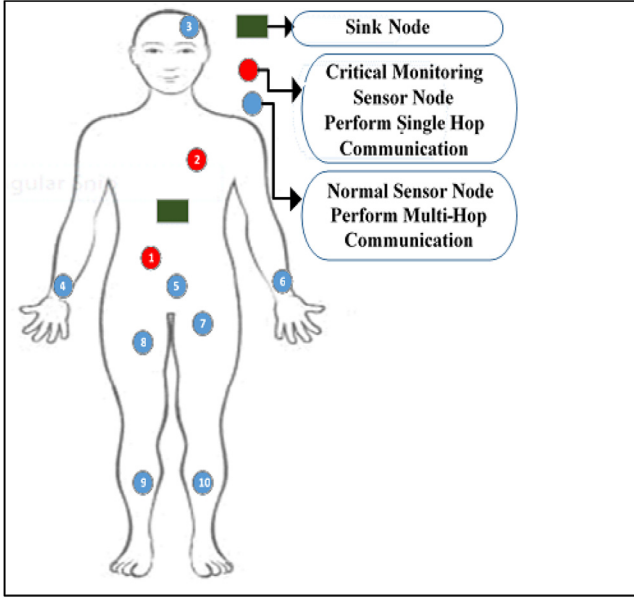


Fig. 3. Nodes deployment.

3.3. System model and network specifications

The system model and network specifications show the type of sensor-nodes deployed in our WBAN and the type of communication that takes place in ERRS. In ERRS, the following two types of homogeneous sensor-nodes with one heterogeneous node are deployed on the human body for optimal monitoring, figuratively shown in Fig. 3.:

Homogeneous Nodes:

- **Critical Monitoring Bio Sensor-Nodes** — Sensor Node (SN)₁ is deployed for Glucose Monitoring, and SN ₂ is deployed for ECG. Both sensors are critical monitoring biosensors that perform single-hop communication and which are placed near the *sink* so that they send their data directly to the sink due to its critical nature.
- **Normal Monitoring Bio Sensor-Nodes** — All other deployed biosensor nodes from SN ₃ to SN ₁₀. These normal monitoring nodes transmit their data through a forwarder node (FVN) to the *sink*. The rationale behind single-hop and multi-hop communication will be discussed later.

Heterogeneous Node:

- **Sink Node** — The sink node receives data from both critical monitoring sensor-nodes and normal monitoring sensor-nodes and sends it to a remote control and analysis center.

3.4. ERRS: Phase wise operations

In this section, we explain the phase-wise operations of the proposed ERRS in detail.

3.4.1. Phase – 1: Network deployment and initialization

Initially, when sensor-nodes are deployed on the human body, as shown in Fig. 3, the network initialization process begins and sensor-nodes are prepared for clustering organization through the following steps:

- (a) The sink node broadcasts a HELLO message containing its location, its neighbor locations, and information about possible routes to the sink.

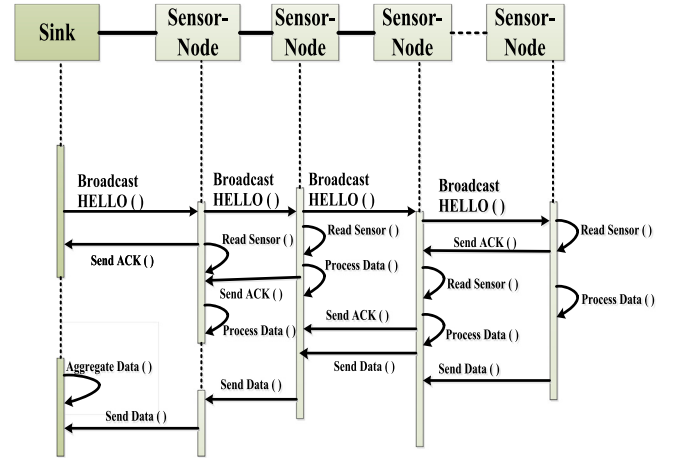


Fig. 4. Sequence diagram of the initialization phase.

- (b) A sensor-node which receives the HELLO packet updates its routing table accordingly and acknowledges the reception with an Acknowledgment Packet to the sink node containing information about its ID (SN_{ID}), its residual energy (SN_{RE}) and location (SN_{LOC}).
- (c) The sink node updates its Information Table accordingly, storing the values of all deployed sensor-nodes who have acknowledged the HELLO message.

The sequence diagram in Fig. 4 demonstrates the above network deployment and initialization phase.

3.4.2. Phase-2: Forwarder node selection

Subsequently to the above, in Phase-2, an appropriate forwarder node is selected by the sink node to organize sensor-nodes into clusters so that the data gathering process is started. The proposed *forwarder node* (FVN) selection process takes place as follows:

- a. During the network deployment and initialization phase, the sink node updates its Information Table, which contains the SN identifier, its location, and current energy status, i.e., SN_{ID} , SN_{Loc} , and SN_{RE} . This information will help to select a suitable forwarder node.
- b. The sink node then computes a weight value of the deployed sensor-nodes using a Minimum Weight Factor (Min_WF) based on the current residual energy and distance of the sensors. From the message exchanged with the sink node, the distance between a sensor and the sink is computed based on the received signal strength. **Weight Factor** — it is a parameter which is a function of the distance of a sensor-node from the sink and its residual energy. Let say $Min_WF(i)$ is the weight factor, $SN_D(i)$ is the distance between the sink and a sensor-node i , and $SN_RE(i)$ is its residual energy, the minimum weight factor can be calculated as follows:

$$Min_WF(i) = SN_D(i) / SN_E_i^{initial} - SN_E_i^{consumed} \quad (1)$$

- c. Using the above weight factor, a sink node then selects a sensor-node with minimum weight factor value as a Forwarder Node.
- d. The ID of the selected Forwarder is then broadcasted to all the deployed sensor-nodes.
- e. Sensor-nodes acknowledge this message with a join request to register with the forwarder node to become its member and get ready for data transmission.
- f. The forwarder node then receives the data from the member nodes and further sends it to the sink node. However, if the distance of the sensor-node to the sink is less than the distance

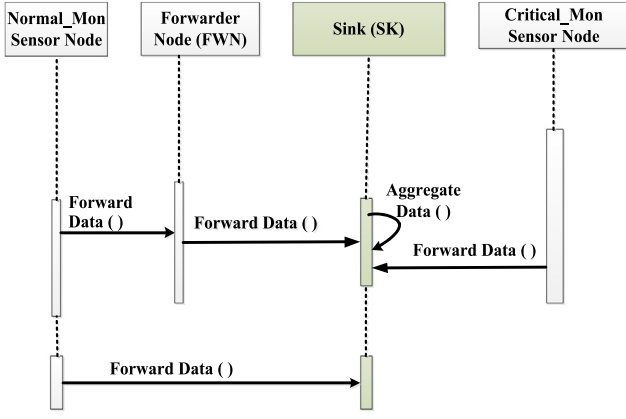


Fig. 5. Data transmission phase.

of the sensor-node to the forwarder node, in that case, the node will transmit its data directly to the sink to avoid communication and processing overhead.

After that, the data transmission continues until the selected forwarder node reaches the pre-defined threshold value, and then another forwarder node is selected based on *Minimum Weight Factor* ($Min_WF(i)$) and so forth. The algorithm (pseudo code) of the forwarder node selection process is as under:

3.4.3. Phase-3: Data transmission

Once the forwarder node is selected and the network is organized, the data transmission phase initiates. In this phase, the normal monitoring sensor-nodes transmit their data to the sink through the forwarder node or directly in some cases, as discussed above in Section 3.4.2 Phase – 2: Forwarder Node Selection. While the critical monitoring sensor-nodes (ECG and GLS Sensors) transmit their data directly to the sink due to the nature of their critical data to avoid any loss or interference in data, as discussed in Fig. 5. For data transmission in WBANs, the most efficient method used by many protocols and algorithms is to schedule the data transmission using the TDMA (Time Division Multiple Access) approaches. In TDMA, the forwarder node will assign time-based slots to its member nodes. Consequently, the member nodes will transmit their data to the forwarder node in their given assigned time slots to minimize the energy consumption through proper scheduling of sensor-nodes [11,27].

The forwarder node then collects the data and sends it to the sink. The sink will monitor the energy status of the forwarder node. For example, during the data transmission process, if the value of the weight factor of the forwarder node reaches a predefined threshold, let say Ω , then the sink will not let the forwarder node take part further in the data forwarding of other sensor-nodes to preserve its energy. Once the threshold value is met, the role of the forwarder node is rotated to normal status (which will be done by the sink) in order to balance the energy load among the deployed sensor-nodes, thus providing an opportunity for all sensor-nodes to have approximately the same lifetime, hence extend stability period and lifetime of the network. The proposed forwarder node rotation technique for load-balancing is explained in the subsequent section.

3.4.4. Phase-4: Rotation of forwarder nodes for load balancing

In Phase-4 of ERRS, a new technique is proposed to rotate the role of the selected forwarder node among the deployed sensor-nodes for load balancing. A new forwarder node is selected by the sink after several data transmission rounds based on the pre-defined threshold value. While the new selection is based on a threshold value Ω of the weight factor of the previous forwarder node, which is a function of

the Residual Energy (RE) and Packet Handling Rate (PHR), as shown in Eq. (2).

$$WF(FWN) = FWN_RE * FWN_PHR \quad (2)$$

Calculation of Packet Handling Rate (PHR) of the Forwarder Node

To calculate the PHR of the forwarder node, most of the required information is taken from the routing table. Moreover, every forwarder node uses *Sent_Array* and *Received_Array* to calculate the PHR.

- (a) **Sent_Array** — Every forwarder node can use *Sent_Array* to keep the record of a total number of sent packets. *Sent_Array* is a two-dimensional array which consists of $N - 1$ rows and two columns, wherever a total number of sensor-nodes in the cluster is represented by N . The ID of neighbor sensor-nodes whose packets are forwarded towards sink by forwarder node contains in column 1, while a total number of sent packets is shown in column 2. Similarly, *Sent_Array* can be mathematically shown in Eq. (3).

$$\sum_k^N SENT(x) [\check{k}, 2] \quad (3)$$

Where x = ID of forwarder node, N = total number of sensors in a cluster, $\check{k}$ = packets sent by forwarder node.

- (b) **Received_Array** — Every forwarder node can use *Received_Array* to hold the record of a total number of received packets. *Received_Array* is also a two-dimensional array that includes a total number of packets received by forwarder node from their member sensor nodes by using TDMA (Time Division Multiple Access) approach.

Mathematically, it can be expressed from the below equation.

$$\sum_{k=1}^N RECEIVED(\check{k}, x) \quad (4)$$

Wherever N = Total number of sensor-nodes in a cluster, x = Sensor-node ID, Total number of rows = $N - 1$, Total number of columns = 2. Furthermore, the ID of neighbor sensor-nodes from where the packets are received is contained in column 1, while the total number of received packets by the forwarder node is obtained from column 2.

Therefore, PHR for forwarder node can be calculated by the below equation.

$$PHR = \sum_{k=1}^N RECEIVED(\check{k}, x) / \sum_{k=1}^N SENT(x) [\check{k}, 2] * 100 \quad (5)$$

This rotation of the role of a forwarder node for load-balancing has the intrinsic advantages of distributing the load uniformly among all the nodes. Every node gets a chance to become a forwarder node and as a result, the energy consumption is more uniform, thereby increasing the stability period and reliability with maximizing throughput. The forwarder node rotation process is as follows:

- The sink node keeps updating the Information Table during the data transmission process and checks the threshold value Ω for the selected forwarder node.
- During the data transmission process, the sink node also computes the New Weight Factor (WF) value of the forwarder node and compares it with predefined threshold value Ω . The WF is a function of the residual energy and Packet Handling Rate of the current forwarder node.
- Once the threshold value Ω for any forwarder node is met, the data transmission process is halted and the forwarder node rotation process is initiated.

Pseudo-code: Forwarder Node Selection Process

Notations:

SNs: Sensor-nodes*SK*: Sink Node*SN_RE*: Sensor Node Residual Energy*FWN*: Forwarder Node*Msg*: Data Message/Control Message

Network Initialization Phase; // Network initialization phase starts after the network deployment

1. START → Initialize for all SNs
2. SK Broadcast Hello_Msg; //Contain position of the sink, Neighbor information and Possible routes to sink
3. Received by SNs = {SN1... SN10}
4. IF SNs in Range & Receive Hello_Msg;
5. THEN Acknowledge (ACK) with SN → SN_ACK
6. SK Receives SN_ACK ∀ SNs & SELECT FWN $\Leftarrow$ Min_Weight;
7. SELECT FWN $\in$ Table: Min_Weight = $SN_D(i) / SN_Ei^{initial} - SN_Ei^{consumed}$

- d. Once a new forwarder node is selected, the sink node broadcasts the ID of the selected Forwarder node to all the deployed nodes and the sensor-nodes acknowledge this message with a join request to register with the forwarder node in order to become its member and get ready for data transmission.
- e. This data transmission phase continues until the selected forwarder node reaches the pre-defined threshold value, and then another forwarder node is selected based on *Minimum Weight Factor (Min_WF(i))* and so on. The pseudo-code of the load-balancing process is given below, while the whole process is illustrated with the help of a flow-chart in Fig. 6.

4. Simulation results and analysis

To evaluate the performance of the proposed scheme and compare it with other existing solutions – in our research for implementation – we have used the MATLAB simulator. MATLAB is an easy to use and flexible simulator, having built-in analysis capabilities with a powerful graphical and interactive environment and which is best suited for the evaluation of WBANs protocols and applications [38].

Following are some of the major evaluation and testing targets that we aimed to achieve from our simulation:

1. Firstly, to perform an experiment in order to check the network stability period and overall network lifetime. The goal of this experiment is to check if our proposed solution can achieve a higher network lifetime and enhanced stability period.
2. Secondly, to perform experiments to check the network throughput in terms of packets send to the sink and packets dropped. We will examine the number of packets received and the total number of dropped packets out of a total number of sent packets during the data transmission phase to evaluate the reliability in terms of network throughput.
3. Thirdly, to calculate and analyze the total energy consumption during/after the data gathering and transmission phase. The aim is to check whether the energy consumption is uniform or uneven.
4. Finally, to evaluate the delay in the network. Delay is the time needed by a packet to reach from source node to the destination. This experiment aims to judge the reliability of the network in terms of timely delivery of data to the sink.

4.1. Assumptions and node deployment

The proposed ERRS is based on the system model and specifications that consist of a set of homogeneous sensor-nodes and a sink node deployed on the human body, as shown in Fig. 7. Static Sensor-nodes on a human body are deployed in the pre-defined locations for adequate monitoring of the patient's body, for instance, Sensor 1 for glucose monitoring and sensor 2 for ECG, are critical monitoring sensors which have direct single-hop communication with the sink due to the critical nature of the data it senses. All other attached sensor-nodes transmit their data to the sink through forwarder nodes continuously. For the implementation of ERRS, some basic assumptions are necessary in order to provide a consistent framework for its operation and to analyze the proposed scheme effectively.

Basic Assumptions:

- The deployed WBAN consists of N nodes, each node having the routing forwarding function.
- All sensor-nodes (homogeneous) have the same communication range except the sink node (heterogeneous).
- A sensor-node can link directly with all the sensor-nodes which are in its communication range.

Network Topology:

The proposed ERRS is based on the clustering hierarchical management architecture [28], and therefore, it is structured as a tree topology, as illustrated in Fig. 7. The snap-shot of the tree topology created with the MATLAB simulator is shown in Fig. 8.

4.2. Simulation parameters and radio energy model

We used the simulated network setup in MATLAB to compare the efficacy of the proposed ERRS in comparison to SIMPLE and M-ATTEMPT protocols [23,27]. In the proposed ERRS scheme, ten sensor-nodes are deployed on the human body in the $X * Y$ network area. Furthermore, two types of homogeneous sensor-nodes (*critical monitoring sensor-node and normal monitoring sensor-nodes*) with one heterogeneous node (*sink node*) are deployed on the human body for optimal monitoring, as shown in Figs. 3 and 7. Sink is positioned at the center of the network, i.e., on the chest of the human body. Throughout the network, initial energy ($SN_Ei^{initial}$) of 1.0 J (1000 mJ) is assigned to each sensor

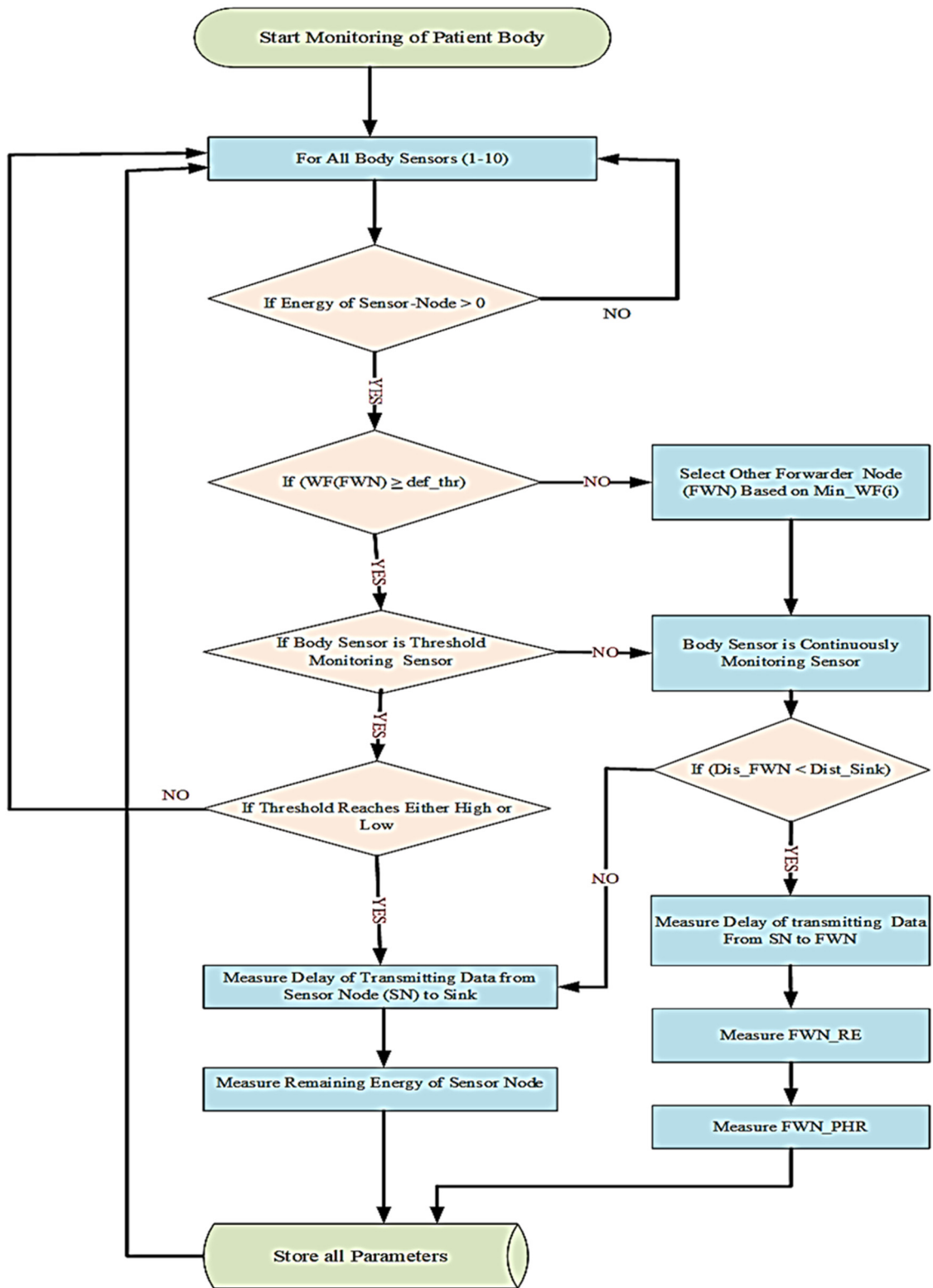


Fig. 6. Flow-chart of the overall operations of the proposed scheme.

node, “ Ω ” represents the threshold value of Weight Factor of current forwarder node, the amount of transmission energy (E_{TXproc}) is 16.7 nJ/bit and transmit amplifier energy (E_{TXproc}) is 1.97 nJ/bit/m², as given in Table 1. Moreover, according to the nature and size of the sensed data, the packet size (k) can be different for different applications.

In WBANs, Nordic nRF 2401A and Chipcon CC 2420 wireless radio transceivers are generally used in [15,18,33]. Table 2 shows a comparison of different parameters of these two radio transceivers. We have used the energy parameters of the Nordic nRF 2401A transceiver, which is a single chip, low power transceiver, because it consumes less power as compared to Chipcon CC2420 as shown in Table 1

Pseudo-code: Load Balancing Process of ERRS Scheme**Notations:***FWN_RE*: Residual Energy of Forwarder Node*Spec_rounds*: Specified Rounds*def_thr*: Pre-defined Threshold*D_{SN(i)}*: Distance of Sensor Node from Sink Node*Dist_FWN*: Distance of Forwarder Node from Sink Node*PHR*: Packet Handling Rate*SN(i)*, *p* = 2: Critical Monitoring Sensors*SK*: Sink Node

```

1. while data aggregation at (FWN) == true do
2.   for i = 1: 1: Node do
3.     if (DSN(i) < Dist_FWN || SN(i). P == 2) then
4.       // Single-Hop Communication
5.       Increment: packet_to_SK = packet_to_SK+1
6.       DSN(i) = sqrt ((S(i). Xd - SK.X) ^2 + (S(i). Yd - SK. Y) ^2);
7.       Select FWN for Spec_rounds ∈ Table ⇔ Min_Weight
8.       if (WF (FWN) >= def_thr) then
9.         Increment: packet_to_FWN = packet_to_FWN+1
10.      Dist_FWN = sqrt ((S(FWN). Xd - (SK.X)) ^2 + (S(FWN). Yd - (SK.Y)) ^2);
11.     end if
12.   end if
13. end for
14. end while

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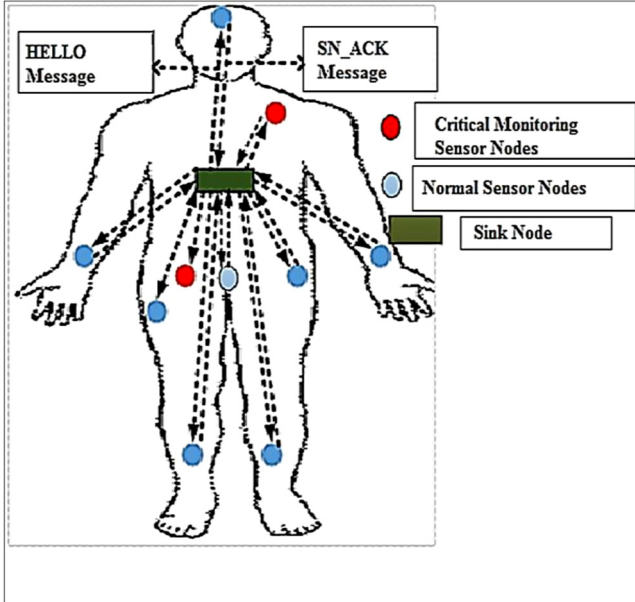


Fig. 7. Network deployment.

There are different radio models used in the literature. However, we have used a first-order radio model to check the performances of various schemes in WBANs [11,15]. This model gives mathematical equations for both data transmission and reception, which provides simplicity in implementation [39]. In WBANs, the transmission distance

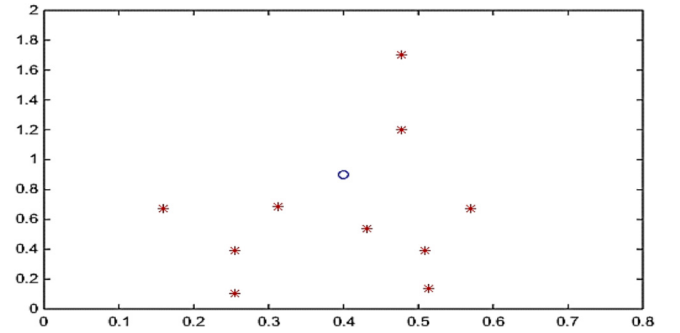


Fig. 8. Network topology creation.

is small and the transceivers may be in the line of sight of each other while assuming that there are no large obstacles or shadowing. Furthermore, this model is based on energy loss due to the square of the distance d^2 between the transmitting and receiving node; therefore, it is best suited for the proposed ERRS as it also accounts distance as the main factor for the calculation of weight factor of forwarder node selection. The energy expenditure during the transmission and reception of energy to process k number of bits over a distance d as:

$$E_{TX}(\check{k}, d) = E_{TXproc}(\check{k}) + \epsilon_{amp}(\check{k}, d) \quad (6)$$

$$E_{TX}(\check{k}, d) = E_{TXproc} \cdot \check{k} + \epsilon_{amp} \cdot \check{k} \cdot d^2 \quad (7)$$

$$E_{RX}(\check{k}) = E_{RXproc}(\check{k}) = E_{RXproc} \quad (8)$$

Table 1

Simulation parameters.

Parameters	Values	Units
Frequency	2.4	GHz
Number of sensor nodes	10	–
Total number of rounds	1000	–
DC current (T_X)	10.5	mA
DC current (R_X)	18	mA
Min. supply voltage	1.9	V
E_{TXproc}	16.7	nJ/bit
E_{RXproc}	36.1	nJ/bit
Energy Consumed in Aggregation (Eda)	5	nJ/bit
ϵ_{amp}	1.97	nJ/bit/m2
Ω	30	%
Packet size ($\check{k}$) (Data + controls information)	7/56	Bytes/Bits
Initial energy (SN _{Eo})	1.0/1000	J/mJ

Table 2

Different parameters of radio transceivers in WBANs.

Parameters	CC2420	nRF 2401A	Units
DC current (T_X)	17.4	10.5	mA
DC current (R_X)	19.7	18	mA
Min. supply voltage	2.1	1.9	V
E_{TXproc}	96.9	16.7	nJ/bit
E_{RXproc}	172.8	36.1	nJ/bit
ϵ_{amp}	271	1.97	nJ/bit/mn

In these equations, E_{TX} is the total energy consumption to transmit a packet, contains k no of bits (in Joules) over a separation distance d . E_{RX} is the amount of energy consumed by the receiver. E_{TXproc} and E_{RXproc} are the energies required to run the electronic circuit of transmitter and receiver, respectively. ϵ_{amp} is the energy required by the amplifier circuit, while d^2 represents the loss of energy due to channel transmission. In WBANs, the communication medium is the human body, which contributes attenuation to the radio signals. Therefore, a path loss coefficient parameter n is included in the radio model. The equation for the transmitter energy consumption is [11,28]:

$$E_{TX}(\check{k}, d) = E_{TXproc} \cdot \check{k} + \epsilon_{amp} \cdot \check{k} \cdot d^n \quad (9)$$

4.3. Performance matrices and experimental results

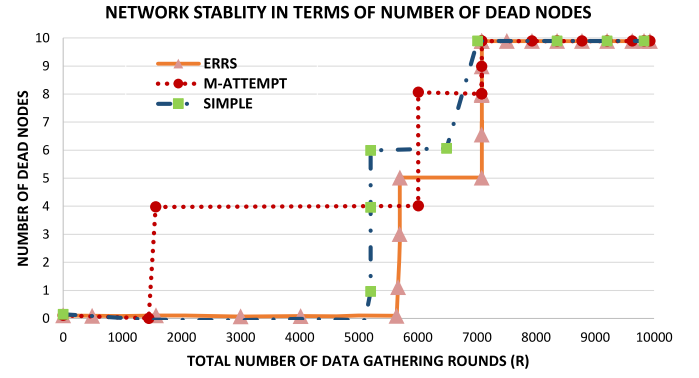
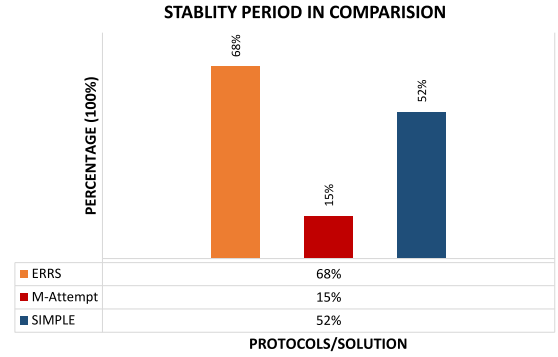
We carried out several sets of experiments using the following key performance matrices:

- Network Stability Period
- Network Throughput
- Residual Energy
- End-to-End Delay

Using these parameters, simulation-based experiments were conducted to evaluate and compare the performance of the proposed ERRS with the benchmark schemes, i.e., SIMPLE and M-ATTEMPT routing protocols to observe ERRS efficacy.

4.3.1. Network stability period

Stability Period in WBAN is the period of network operation from the start of the network until the first node in the network dies. Once the first node dies, the remaining operational time is known as an unstable period. The stability period in the network is highly affected by the traffic load distribution and the selection criteria of the forwarder nodes [11]. In a cluster-based paradigm, it is always desirable to select a forwarder node having higher residual energy, and least distance to the sink node. Furthermore, the stability period has a direct impact on the network operational lifetime. If the stability period is increased, the overall network operational lifetime will be maximized.

**Fig. 9.** Stability period vs. number of dead nodes.**Fig. 10.** Comparison of the stability period of ERRS, M-ATTEMPT and SIMPLE.

Experiment 1 — In this first experiment — result shown in Fig. 9, the stability period in terms of network lifetime of the ERRS scheme is evaluated for 10 sensor-nodes deployed with one heterogeneous sink node and for 10000 data gathering rounds with the network parameters and setup described above in Table 1. It is shown in the graph that the proposed ERRS performed better than those of SIMPLE and M-ATTEMPT protocols. ERRS has an increased stability period, lasting until almost 6000 data gathering rounds. The superior performance of ERRS is due to the selection of a forwarder node after a specified data gathering rounds based on the threshold value of the new weight factor of forwarder node, i.e., WF . In this way, every sensor-node has a chance to become a forwarder node, and the energy load is evenly distributed among the deployed sensor nodes. As a result, ERRS avoids excessive use of a sensor-node's energy and prevents nodes from early death and energy holes. As a result, the stability period is increased, and the network lifetime is maximized. The inferior performance of M-ATTEMPT and SIMPLE protocol is due to the reason that sensor-nodes nearer to the sink node have more chances to become a forwarder node again and again. Hence, the residual energy of that node depletes and quickly dies, which creates energy holes and affects the stability period and network lifetime.

Similarly, Fig. 10 shows the comparison of ERRS, M-ATTEMPT and SIMPLE protocol. The stability period of the M-ATTEMPT is the lowest comparatively because M-ATTEMPT uses the minimum hop-count strategy to route the packets, once the temperature of the sensor-node reaches a pre-defined threshold value, the node does not forward the packets further for other nodes. Therefore, an alternate route is established, which in most cases, is the longer route; hence a sensor-node spends more energy in the transmission of packets on longer routes and ultimately dies quickly.

4.3.2. Network throughput

Network throughput is the total number of packets successfully received at the sink node [30]. WBAN has critical and vital patients'

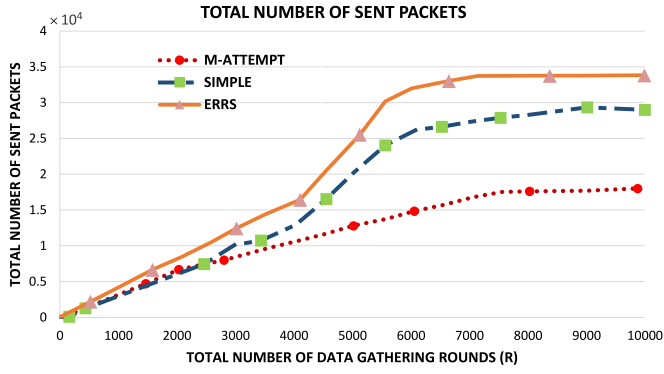


Fig. 11. Total No. of packets sent to sink.

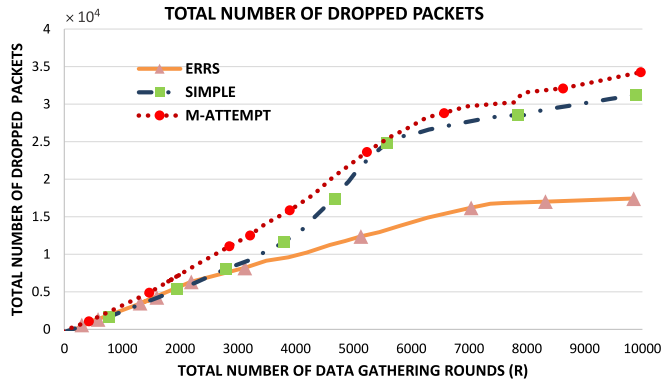


Fig. 12. Packets dropped ratio.

data, so it needs a protocol that has the minimum number of packets drop ratio and maximum reception of data at the sink.

Experiment 1 — In this experiment, we evaluated the network throughput of our proposed ERRS and proved that ERRS achieves maximum throughput comparatively, as shown in Fig. 11. For evaluation and analysis, we considered the same simulation parameters, as discussed in Section 4.2 *Simulation Parameters and Radio Energy Model* above. Therefore, for 1000 data gathering round of 10 deployed sensor-nodes will yield a maximum throughput of 10,000 packets transferred, and subsequently, in 10000 data gathering rounds, 1,00,000 packets are delivered to the sink. Since the packets delivery to the sink node depends on the number of active or alive nodes; therefore, more alive nodes in the network will yield to deliver more data packets to the sink, ultimately maximizing the network throughput. The maximum throughput of the ERRS scheme is due to the increased stability period as compared to the M-ATTEMPT and SIMPLE protocol. However, the throughput of the network may vary for different data gathering patterns, e.g., the throughput of a *time-driven* and *event-driven* application may be different [10].

Experiment 2 — In this experiment, we calculate the total number of dropped packets out of the total number of sent packets to sink, as shown in Fig. 12. Packets dropped is directly proportional to the total number of transmissions, which is calculated based on the *Random Uniformed Model* [11].

Fig. 12 illustrates that the number of dropped packets in ERRS is lower as compared to M-ATTEMPT and SIMPLE protocols. Because the stability period of ERRS is longer than the stability period of M-ATTEMPT, and SIMPLE protocol.

Experiment 3 — In this experiment, we calculate the total number of data packets successfully received at the sink. The maximum data reception at the sink with minimum packets dropped is the design goal of every WBAN's application, which has a significant impact on

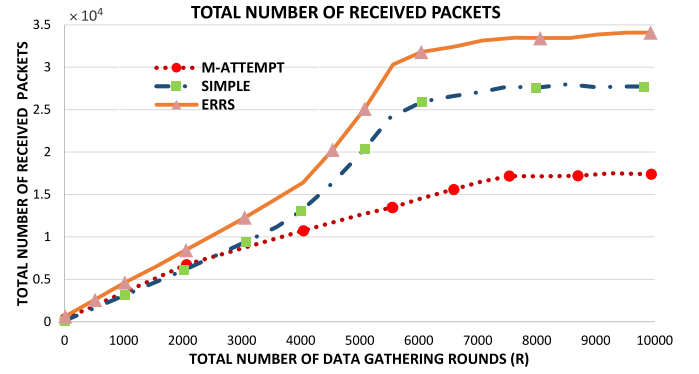


Fig. 13. Total No. of packets received at sink.

the reliability of the network. Therefore, for a reliable WBAN, it is required that the sensed data should be correctly received at the sink. Mathematically, packet delivery ratio (PDR) in terms of throughput can be calculated, as shown in Eq. (10) [40].

$$PDR = Q_{RP}/Q_{GP} \quad (10)$$

Where Q_{RP} is the amount of data packets received by the sink node and Q_{GP} represents the amount of data packets generated by each sensor-node. The graph in Fig. 13. shows that ERRS has higher throughput as compared to M-ATTEMPT and SIMPLE protocols in terms of data packets received at the sink node.

The overall throughput in terms of the total number of packets received at the sink, ERRS shows an improvement of 23% and 72% than by SIMPLE and M-ATTEMPT protocols, respectively. The higher throughput is due to the reduced data packets loss in the network, that is because the stability period is longer due to the uniform load-distribution in ERRS.

4.3.3. Residual energy/energy consumption

Since sensor-nodes in WBANs are battery-powered having limited energy; therefore, residual energy of a sensor-node is an important aspect that needs to be considered in the design and development of WBANs. Given that, the analysis of the residual energy parameter is important because the energy consumption of a single node has a direct impact on the stability period and reliability of the network. Furthermore, the residual energy of a particular sensor-node is also crucial, because in the proposed ERRS, the residual energy of sensor-node to become a potential new forwarder node is considered in the rotation/load-balancing phase as discussed in Section 3.4.4 Phase – 4: Rotation of Forwarder Nodes for Load Balancing, Phase-4. In this experiment, we evaluated the residual energy of deployed sensor-nodes in terms of energy consumption during the defined data gathering/transmission round (the Phase-3 of the proposed ERRS). The energy-consumption analysis of a sensor-node using a single-hop and multi-hop communication by using the First Order Radio Model is as follows:

Energy consumption for single-hop communication is:

$$E_{SH} = E_{TX} \quad (11)$$

E_{TX} is the transmission energy, presented as Eq. (12) below:

$$E_{TX} = \check{\kappa} \times (E_{TXproc} + \epsilon_{amp}) \times d^2 \quad (12)$$

Where E_{TXproc} is the energy consumed by the electronic circuit.

Now the energy consumed during multi-hop communication is given by, Eq. (13):

$$E_{MH} = \check{\kappa}(\bar{n} \times (E_{TX}) + (\bar{n} - 1) \times (E_{RX} + E_{da})) \quad (13)$$

E_{RX} is the reception energy and $\bar{n}$ is the number of nodes.

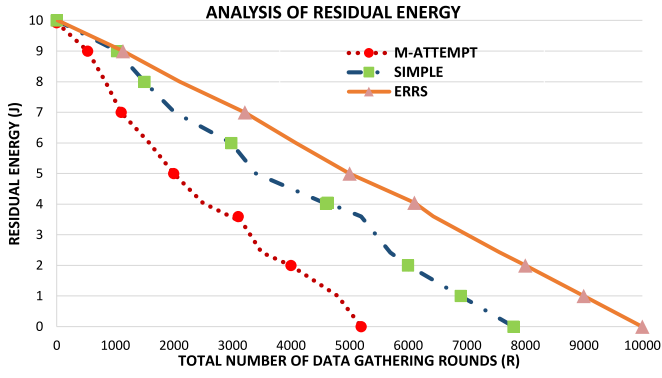


Fig. 14. Analysis of residual energy.

Furthermore, the transmission energy of a forwarder node can be calculated as using Eq. (14):

$$E_{TX}^{FWN} = \tilde{n} \cdot E_{TX}(\tilde{k} \cdot d) + \tilde{n} \cdot \tilde{k} \cdot E_{TXproc} \quad (14)$$

Where $\tilde{n}$ is the over-all number of member nodes and $\tilde{k}$ is packet length or size. d shows a distance of forwarder nodes from the sink. Furthermore, E_{TX} and E_{RX} represent the energy consumption costs of the transmitter and receiver per packet. Therefore, E_{TXproc} and E_{RXproc} are the energy required to run the electronic circuits of the transmitter and receiver. Consequently, in terms of complete network lifetime, the total transmit energy of a forwarder node can simply be calculated as shown in Eq. (15) [28].

$$E_{TX}^{NLT} = \int^t E_{TX}^{FWN} dt \quad (15)$$

Fig. 14, shows the average remaining energy of the 10 nodes in 10000 data gathering rounds. Since each node has 1 Joule of initial energy, therefore, the total energy of the network is 10 Joules. Simulation is run till the 10000 data gathering rounds are completed and when all the deployed sensor-nodes exhaust their residual energy. From the graph in Fig. 14, it can be observed that the energy consumption of the proposed ERRS is linear as compare to the M-ATTEMPT protocol. While SIMPLE and ERRS, both have competitive energy consumption. However, during the 5000–6000 data gathering rounds, in the SIMPLE protocol, the residual energy of sensor-nodes drastically reduces, and around 7000–8000 rounds, all nodes expend their residual energy.

However, ERRS performs superior to M-ATTEMPT and SIMPLE protocols, the residual energy of sensor-nodes lasts till 10000 data gathering rounds, ultimately extends the network lifetime. The better performance of ERRS is due to the use of an adaptive hierarchical clustered based routing mechanism, where a farther node transfers its data to sink through a *forwarder node*. Furthermore, a *forwarder node* in ERRS is selected using a cost function, which is more energy-efficient comparatively, as explained in Section 3.4.2 Phase – 2: Forwarder Node Selection. This efficient selection of *forwarder node* and appropriate route selection plays a significant role in the energy conservation of the deployed sensor-nodes of the proposed scheme. Besides, due to the rotation of the forwarder node, the energy load is uniformly distributed, which prevents the exhaustion of the sensor-nodes energy and energy holes are avoided; consequently, the network lifetime is extended.

4.3.4. End-to-end delay

End-to-End Delay is the time needed by a packet to reach from source node to the destination (i.e., from a sensor-node to the sink). In WBANs, sensor-node monitors various physical phenomena, e.g., the Physiological sensors measure critical physical attributes such as blood pressure, temperature, and monitoring of glucose, etc. [7]. Therefore, a delay must be examined to ensure the timely delivery of the critical data to the sink in medical applications. End-to-End Delay (EED) metric

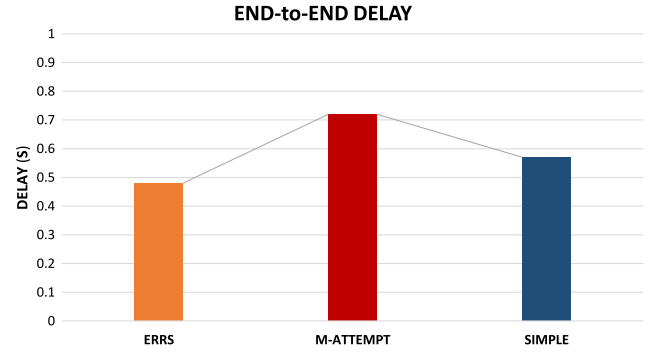


Fig. 15. End-to-end delay.

is calculated by subtracting the time at which the first packet was transmitted by a source from the time at which that first data packet arrived at the destination [41,42].

Mathematically, the overall average EED can be calculated as given in Eq. (16), [43].

$$Avg. EED = TT_{SP} / TRP \quad (16)$$

Where TT_{SP} is the total time required to send packets and TRP is the total number of packets received by all destination nodes.

Experiment — In this experiment, an average EED is calculated, as shown in the graph in Fig. 15. We have used the same experimental setup, as described in Section 4.2 *Simulation Parameters and Radio Energy Model*. For the sake of experimentation, let us consider that the patient/human body is one meter² in length in XY-plane. Therefore, there are 10 sensor-nodes deployed on the body on different locations and a data packet of 7 Bytes is transferred from the sensor-node to the sink. Moreover, we assumed that, on average, a packet travels at a speed of 1 m/s using a typical wireless medium and setup. Given these parameters, an average end-to-end delay of ERRS, M-ATTEMPT, and SIMPLE was calculated. It is evident from the graph that the proposed ERRS scheme incurs less delay as compared to M-ATTEMPT and SIMPLE protocols. The minimized delay in packet transmission is due to the use of an adaptive static clustering approach in the proposed ERRS. On the other hand, in M-ATTEMPT, sensor-nodes send their data to a forwarder node that could be located at a long distance, and it is a known fact that the distance has a direct impact on the delay as delay increases with the increase in the distance [43]. Hence, in M-ATTEMPT, the higher delay is due to longer distances of some nodes from the sink. While SIMPLE protocol exhibits higher end-to-end delay due to the frequent rotation of forwarder node after each data gathering round. To conclude, the end-to-end delay of the proposed ERRS is improved by 17% and 40% than by SIMPLE, and M-ATTEMPT respectively.

5. Conclusion and future work

In this paper, we have proposed a new energy-efficient and reliable routing technique to enhance the stability period in WBAN known as ERRS. ERRS comprised two novel solutions, namely, the Forwarder Node Selection, and Forwarder Node Rotation techniques. The proposed ERRS takes advantage of the adaptive static clustering routing technique and achieves enhanced stability period, and longer network lifetime ultimately maximizes reliability. Through simulation results based on the MATLAB tool, ERRS showed an improvement of 26% over the SIMPLE protocol in terms of the network stability period and throughput. Moreover, the end-to-end delay of the proposed ERRS is improved by 17% and 40% than by SIMPLE and M-ATTEMPT protocols, respectively, which proves ERRS to be an efficient and reliable routing solution for WBANs.

The proposed ERRS has been proved to be an efficient and reliable routing solution for WBANs. However, to deal with the broad range

of possible applications for WBANs, related to this, a key issue of scalability and mobility will be considered in our future work. For instance, it will be interesting to investigate further the implications of the mobility of different human body parts on the overall stability and network lifetime of WBANs. Similarly, the security aspect of the proposed scheme can be tested in the future for life-critical applications, which requires more accuracy in terms of reliability and latency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] S. Movassaghi, M. Abolhasan, J. Lipman, D. Smith, A. Jamalipour, *Wireless body area networks: A survey*, *IEEE Commun. Surv. Tutor.* 16 (2014) 1658–1686.
- [2] A. Rahim, N.C. Karmakar, Sensor cooperation in wireless body area network using network coding for sleep apnoea monitoring system, in: *Intelligent Sensors, Sensor Networks and Information Processing*, 2013 IEEE Eighth International Conference on, 2013, pp. 432–436.
- [3] V.M. Birari, J. Helonde, V.M. Wadhai, *Algorithmic approach for reliable communication in wireless body area network for patient monitoring system*, *Int. J. Eng. Econ. Manag.* 2 (2014) 1–5.
- [4] X. Lai, Q. Liu, X. Wei, W. Wang, G. Zhou, G. Han, *A survey of body sensor networks*, *Sensors* 13 (2013) 5406–5447.
- [5] K. Deepak, A. Babu, *Packet size optimization for energy efficient cooperative wireless body area networks*, in: 2012 Annual IEEE India Conference, INDICON, 2012, pp. 736–741.
- [6] W. Kurschl, S. Mitsch, J. Schoenboeck, *Modeling distributed signal processing applications*, in: 2009 Sixth International Workshop on Wearable and Implantable Body Sensor Networks, 2009, pp. 103–108.
- [7] P.M.P. Dharshini, M. Tamilarasi, *Adaptive reliable cooperative data transmission technique for Wireless Body Area Network*, in: *Information Communication and Embedded Systems, ICICES*, 2014 International Conference on, 2014, pp. 1–4.
- [8] A. Tauqir, N. Javaid, S. Akram, A. Rao, S. Mohammad, *Distance aware relaying energy-efficient: Dare to monitor patients in multi-hop body area sensor networks*, in: *Broadband and Wireless Computing, Communication and Applications, BWCCA*, 2013 Eighth International Conference on, 2013, pp. 206–213.
- [9] V.M. Birari, J. Helonde, V. Wadhai, *Seamless mobility with interference free reliable communication in WBAN*, in: *Computing Communication Control and Automation, ICCUBE*, 2015 International Conference on, 2015, pp. 251–255.
- [10] M.M. Sandhu, M. Akbar, M. Behzad, N. Javaid, Z.A. Khan, U. Qasim, *REEC: Reliable energy efficient critical data routing in wireless body area networks*, in: *Broadband and Wireless Computing, Communication and Applications, BWCCA*, 2014 Ninth International Conference on, 2014, pp. 446–451.
- [11] M.M. Sandhu, N. Javaid, M. Akbar, F. Najeeb, U. Qasim, Z. Khan, *FEEL: Forwarding data energy efficiently with load balancing in wireless body area networks*, in: *Advanced Information Networking and Applications, AINA*, 2014 IEEE 28th International Conference on, 2014, pp. 783–789.
- [12] H.B. Elhadj, L. Chaari, L. Kamoun, *A survey of routing protocols in wireless body area networks for healthcare applications*, *Int. J. E-Health Med. Commun.* 3 (2012) 1–18.
- [13] W. Jiang, Z. Wang, M. Feng, T. Miao, *A survey of thermal-aware routing protocols in wireless body area networks*, in: *Computational Science and Engineering, CSE and Embedded and Ubiquitous Computing, EUC*, 2017 IEEE International Conference on, 2017, pp. 17–21.
- [14] B. Jan, H. Farman, H. Javed, B. Montrucchio, M. Khan, S. Ali, *Energy efficient hierarchical clustering approaches in wireless sensor networks: A survey*, *Wirel. Commun. Mob. Comput.* 2017 (2017).
- [15] M.M. Sandhu, N. Javaid, M. Imran, M. Guizani, Z.A. Khan, U. Qasim, *BEC: A novel routing protocol for balanced energy consumption in Wireless Body Area Networks*, in: 2015 International Wireless Communications and Mobile Computing Conference, IWCMC, 2015, pp. 653–658.
- [16] K. Sarangi, I. Bhattacharya, *A study on data aggregation techniques in wireless sensor network in static and dynamic scenarios*, *Innov. Syst. Softw. Eng.* 15 (2019) 3–16.
- [17] T.S. Panag, J. Dhillon, *Dual head static clustering algorithm for wireless sensor networks*, *AEU-Int. J. Electron. Commun.* 88 (2018) 148–156.
- [18] H.-W. Ferng, J.-S. Chuang, *Area-partitioned clustering and cluster head rotation for wireless sensor networks*, in: 2017 International Conference on Machine Learning and Cybernetics, ICMMLC, 2017, pp. 593–598.
- [19] S. Movassaghi, M. Abolhasan, J. Lipman, *A review of routing protocols in wireless body area networks*, *J. Nets.* 8 (3) (2013) 559–575.
- [20] N. Yessad, M. Omar, A. Tari, A. Bouabdallah, *QoS-based routing in wireless body area networks: A survey and taxonomy*, *Computing* 100 (2018) 245–275.
- [21] J.I. Bangash, A.H. Abdullah, M.H. Anisi, A.W. Khan, *A survey of routing protocols in wireless body sensor networks*, *Sensors* 14 (2014) 1322–1357.
- [22] C.H.W. Oey, S. Moh, *A survey on temperature-aware routing protocols in wireless body sensor networks*, *Sensors* 13 (2013) 9860–9877.
- [23] N. Javaid, Z. Abbas, M. Fareed, Z. Khan, N. Alrajeh, *M-ATTEMPT: A new energy-efficient routing protocol for wireless body area sensor networks*, *Procedia Comput. Sci.* 19 (2013) 224–231.
- [24] A. Ahmad, N. Javaid, U. Qasim, M. Ishfaq, Z.A. Khan, T.A. Alghamdi, *RE-ATTEMPT: A new energy-efficient routing protocol for wireless body area sensor networks*, *Int. J. Distributed Sens. Netw.* 2014 (2014).
- [25] Y. Qu, G. Zheng, H. Ma, X. Wang, B. Ji, H. Wu, *A survey of routing protocols in wban for healthcare applications*, *Sensors* 19 (2019) 1638.
- [26] R. Jafari, M. Effatparvar, *Cooperative routing protocols in wireless body*, *Int. J. Comput. Inf. Technol.* 5 (2017) 43–51.
- [27] Q. Nadeem, N. Javaid, S.N. Mohammad, M. Khan, S. Sarfraz, M. Gull, *Simple: Stable increased-throughput multi-hop protocol for link efficiency in wireless body area networks*, in: 2013 Eighth International Conference on Broadband and Wireless Computing, Communication and Applications, 2013, pp. 221–226.
- [28] N. Javaid, A. Ahmad, Q. Nadeem, M. Imran, N. Haider, *iM-SIMPLE: iMproved stable increased-throughput multi-hop link efficient routing protocol for wireless body area networks*, *Comput. Hum. Behav.* 51 (2015) 1003–1011.
- [29] J. An, D. Sethi, *Comparative analysis of energy efficient routing in WBAN*, in: 2017 3rd International Conference on Computational Intelligence & Communication Technology, CICT, 2017, pp. 1–6.
- [30] S. Ahmed, N. Javaid, M. Akbar, A. Iqbal, Z. Khan, U. Qasim, *LAEEBA: Link aware and energy efficient scheme for body area networks*, in: *Advanced Information Networking and Applications, AINA*, 2014 IEEE 28th International Conference on, 2014, pp. 435–440.
- [31] S. Ahmed, N. Javaid, S. Yousaf, A. Ahmad, M. Sandhu, M. Imran, et al., *Co-LAEEBA: Cooperative link aware and energy efficient protocol for wireless body area networks*, *Comput. Hum. Behav.* 51 (2015) 1205–1215.
- [32] T. Umer, M. Amjad, M.K. Afzal, M. Aslam, *Hybrid Rapid Response Routing Approach for Delay-Sensitive Data in Hospital Body Area Sensor Network*, 2016, pp. 1–7.
- [33] G. Mehmood, M.Z. Khan, S. Abbas, M. Faisal, H.U. Rahman, *An energy-efficient and cooperative fault-tolerant communication approach for wireless body area network*, *IEEE Access* 8 (2020) 69134–69147.
- [34] Z. Ullah, I. Ahmed, K. Razzaq, M.K. Naseer, N. Ahmed, *DSCB: Dual sink approach using clustering in body area network*, *Peer Peer Netw. Appl.* 12 (2019) 357–370.
- [35] M.A. Kumar, C.V. Raj, *On designing lightweight Qos routing protocol for delay-sensitive wireless body area networks*, in: 2017 International Conference on Advances in Computing, Communications and Informatics, ICACCI, 2017, pp. 740–744.
- [36] A. Ghani, H.A. Naqvi, M. Sher, Z.S. Khan, I. Khan, M. Saqlain, *Energy efficient communication in body area networks using collaborative communication in Rayleigh fading channel*, *Telecommun. Syst.* (2015) 1–14.
- [37] H. Wei, H. Li, J. Tan, *Body sensor network based context-aware QRS detection*, *J. Signal Process. Syst.* 67 (2012) 93–103.
- [38] Q.I. Ali, *Simulation framework of wireless sensor network (WSN) using MATLAB/SIMULINK software*, in: *MATLAB-A Fundamental Tool for Scientific Computing and Engineering Applications*, vol. 2, Intech, 2012.
- [39] M. Arshad, M.Y. Aalsalem, F.A. Siddiqui, *Energy efficient cluster head selection in mobile wireless sensor networks*, *J.Eng. Sci. Technol.* 9 (2014) 728–746.
- [40] P. Rohal, R. Dahiya, P. Dahiya, *Study and analysis of throughput, delay and packet delivery ratio in MANET for topology based routing protocols (AODV, DSR and DSDV)*, *Int. J. Adv. Res. Eng. Technol.* 1 (2013) 54–58.
- [41] R. Balaji, P. Raghavendran, R. Ashokan, *Performances enhancement in wireless body area network (WBAN)*, *Int. J. Eng. Innov. Technol.* 3 (2013).
- [42] A. Sangwan, P.P. Bhattacharya, *Delay tolerant energy efficient protocol for inter-BAN communication in mobile body area networks*, *Int. J. Adv. Sci. Eng. Inf. Technol.* 8 (2018).
- [43] S. Pratheema, K.G. Srinivasagan, J. Naskath, *Minimizing end-to-end delay using multipath routing in wireless sensor networks*, *Int. J. Comput. Appl.* 21 (2011) 20–26.